30-DAY GIT LEARNING SERIES

GIT QUEST

LEVEL 6

git diff: Spot the Difference

Week 1: First Steps | Absolute Beginner

QUEST PROGRESS: DAY 6 OF 30

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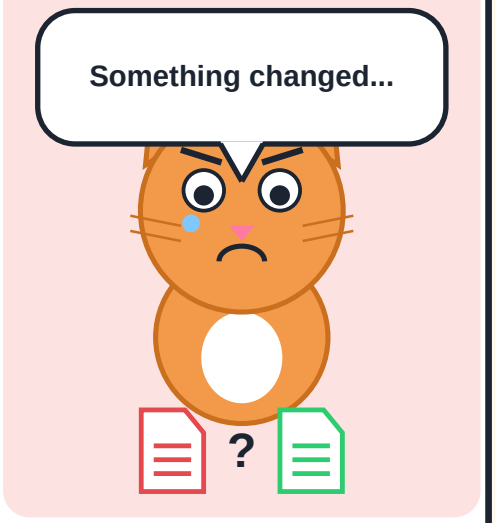
The Story: Red Pen, Green Pen

Commit the Cat knows something changed. But what?

1



The puzzle

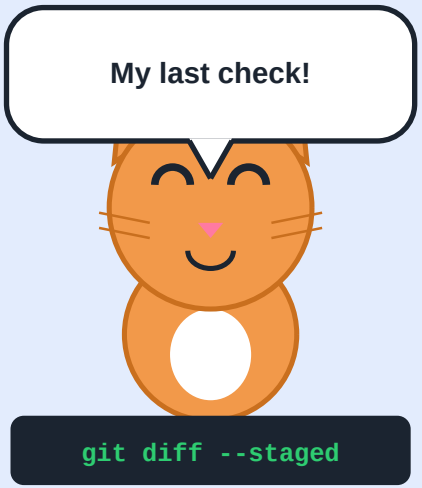
Two versions of a file look almost the same. Which lines are different?

2

git diff

Shows what you changed but have not packed yet. Red is removed, green is added.

3



--staged

Shows what is on the packing table: exactly what your next commit will contain.

4

HEAD~1 HEAD

Compares two commits, so you can see what changed between save points.

Reading a Diff

Red pen = removed. Green pen = added.

```
diff --git a/recipe.txt b/recipe.txt
--- a/recipe.txt
+++ b/recipe.txt
@@ -1,3 +1,4 @@
 Pasta
-Tomato sauce
+Pesto sauce
 Basil
+Cheese
```

1. **`--- and +++`**
   Old version (a/) and new version (b/) of the file.
2. **`@@ -1,3 +1,4 @@`**
   Where: old file showed 3 lines from line 1, new shows 4.
3. **`- (red)`**
   This line was removed.
4. **`+ (green)`**
   This line was added.
5. **`(space)`**
   Unchanged line, shown for context.

WHICH TWO THINGS DOES IT COMPARE?

`git diff`

Desk vs packing table

What did I change but not pack yet?

`git diff --staged`

Packing table vs cupboard

What am I about to commit?

`git diff HEAD`

Desk vs cupboard

Everything since my last save.

`git diff A B`

Two save points

What changed between these commits?

Empty git diff after git add? Try git diff --staged !

Quest Board: Spot the Difference

Finish 4 quests, then collect 3 commits as loot.

1 Save a tiny recipe

```
$ git init -b main spot-diff && cd spot-diff
$ printf "Pasta\nTomato sauce\nBasil\n" > recipe.txt
$ git add recipe.txt && git commit -m "Add recipe"
```

2 Change it, then spot the difference

```
$ printf "Pasta\nPesto sauce\nBasil\nCheese\n" > recipe.txt
$ git diff
```

Red line removed, green line added.

3 Pack it and compare again

```
$ git add recipe.txt
$ git diff              # empty!
$ git diff --staged
```

4 Commit, then compare two saves

```
$ git commit -m "Switch to pesto and add cheese"
$ git diff --stat HEAD~1 HEAD
```

LOOT: TODAY'S 3 COMMITS

COMMIT 1

day06-git-diff/lesson.md

```
docs(day06): add lesson on git diff
```

The lesson: red pen, green pen.

COMMIT 2

day06-git-diff/demo.tape

```
feat(day06): add VHS tape for git diff demo
```

Script that records the terminal GIF.

COMMIT 3

exercise.md + days-1-to-6-summary.md

```
docs(day06): add exercise and days 1-6 summary
```

Challenge, plus a recap of Days 1 to 6.

Tip: make it a habit to run git diff --staged right before every commit.

Boss Fight and Bonus Loot

Cheat sheet, word view, troubleshooting and your next unlock.

BONUS 1: DAY 6 CHEAT SHEET

```
git diff                Desk vs packing table
git diff --staged       Packing table vs cupboard
git diff HEAD           Desk vs cupboard
git diff HEAD~1 HEAD    Last two save points
git diff --stat         Short summary
git diff <file>         One file only
```

BONUS 2: WORD-LEVEL VIEW

```
$ git diff --word-diff
[-Tomato-]{+Pesto+} sauce
```

BONUS 3: WHY IS git diff EMPTY?

```
Changes already staged   -> use git diff --staged
File is new (untracked)  -> git add it first
Nothing changed          -> edit the file
Stuck on (END)           -> press q
```

BOSS FIGHT: QUICK QUIZ

1. Which command shows changes you have already staged?
2. What does a line starting with - mean?
3. Why is git diff empty right after git add?

Answers: 1) git diff --staged 2) The line was removed 3) The changes moved to the packing table

CHALLENGE

Change one line and add one line. Predict what git diff and git diff --staged will show, then check yourself!

LEVEL 7: Week 1 Recap

Unlocks tomorrow. A quiz and a cheat sheet to lock in everything from Days 1 to 6.

LOCKED